Track DC v0.1 — Bell sub-Tsirelson 1/8 explicit derivation framework: 8-sided die candidate (b)

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Track DC v0.1 — Bell 1/8 derivation framework

1. Cal #29 STANDING question-shape audit (applied at design)

Question: "Does substrate dynamics produce the Bell sub-Tsirelson deviation Tsirelson^2 - S^2_BST = 1/2^N_c = 1/8 via an EXPLICIT MECHANISM (not just numerical match)?"

Audit: - Structurally determined? PARTIAL — the *result* 1/8 is fixed by T2399 STRUCTURALLY VERIFIED EXACT identity (substrate-algebra). The *mechanism producing it* is open. - Back-fittable? POTENTIAL RISK — any 8-path decomposition with 1 forbidden path arithmetically gives 7/8, BUT the question asks for substrate-natural identification of the forbidden path with substrate-mechanism (not 1 out of 8 chosen to match the target). - Pre-suppositions? Cal #132 SVC for {Q̂, P̂_op} = 0 (step 1) + Cal #132 FRAMEWORK-PLUS for [B̂, T̂_tick] (step 9 vacuum kicker)

Cal #29 finding: question shape has back-fit risk because the 8-path structure was identified AFTER seeing the 1/8 target. To pass Cal #29, the candidate mechanism (b) must independently derive (1) which 8 paths, (2) which one is structurally forbidden, (3) why the redistribution gives precisely 1/8 (not just approximately).

v0.1 disposition: FRAMEWORK-PLUS, NOT promotion to SVC. Cal #29 risk-flag preserved through derivation work. Explicit substrate-mechanism for each of (1)-(3) is multi-week v0.2+ work.

2. The Bell 1/8 structural identity (T2399 STRUCTURALLY VERIFIED)

Per T2399 (Wednesday May 19): substrate Bell-CHSH operator B̂ on H^2(D_IV^5) has maximum eigenvalue:

B^2_max = (C_2 · N_c · g) / 2^(rank^2) = (6 · 3 · 7)/16 = 126/16 = S^2_BST

Compare to Tsirelson^2 = 8 (qubit Hilbert space Bell-CHSH max):

Tsirelson^2 - S^2_BST = 8 - 126/16 = 128/16 - 126/16 = 2/16 = 1/8 = 1/2^N_c

The 1/8 deviation is EXACT identity at substrate algebra level. Question is the dynamical substrate-mechanism producing it.

2.1 Factor decomposition

128 = 2^7 = 2^g (BST primary g = 7) 126 = 2^g - 2 = C_2 · N_c · g (with C_2 = 6, N_c = 3, g = 7) 2 = 128 - 126 = (2^g - C_2 · N_c · g)

Substantive structural identity: 2^g - C_2 · N_c · g = 2 for BST primaries (g = 7, C_2 = 6, N_c = 3): 128 - 126 = 2 ⍰

This is the substrate-algebraic source of the 1/8 deviation: a 2-unit "gap" between 2^g and C_2 · N_c · g, normalized by 2^(rank^2) = 16.

3. Candidate (b) — 8 = 2^N_c commitment paths via 3 commuting Cartan generators

Per v0.4 Section 6.2 candidate (b): per substrate-tick, the substrate has 2^N_c = 8 *potential* commitment paths corresponding to 3 commuting Cartan generators × {±1} simultaneous eigenvalues.

3.1 Three commuting Cartan generators

A_sub contains a Cartan subalgebra h_Asub including: - T_3^color (SU(3) color Cartan, diagonal generator) - T_8^color (SU(3) color Cartan, second diagonal generator) — but this has eigenvalues in $\{2/\sqrt{3}, -1/\sqrt{3}, -1/\sqrt{3}\}$, not ±1 - T_3^weak (SU(2)_L weak isospin Cartan) - Q̂ (electric charge) - σ_BF (Pin(2) Z_2 sublattice grading) - $\hat{\gamma}^5$ (Dirac chirality, on fermion sublattice)

For the 2^N_c = 8 candidate structure, we need 3 commuting generators with binary ±1 eigenvalues.

Candidate sets:

Set (b.i): {Q̂, σ_BF, $\hat{\gamma}^5$} — but $\hat{\gamma}^5$ is undefined on bosons; restrict to fermion sublattice - 3 binary generators × 2^3 = 8 simultaneous eigenvalue combinations - These don't all commute: {Q̂, P̂_op} = 0 anti-commutation per step 1 - Wait — Q̂ and σ_BF commute? σ_BF is diagonal in K-type; Q̂ is diagonal in K-type; yes [Q̂, σ_BF] = 0 - Q̂ and $\hat{\gamma}^5$: $\hat{\gamma}^5$ acts within fermion sublattice; Q̂ is diagonal; [Q̂, $\hat{\gamma}^5$] = 0 on fermion sublattice

Set (b.ii): {σ_BF, $\hat{\gamma}^5$, T_3^weak} — within fermion sublattice - 3 commuting binary operators - 2^3 = 8 simultaneous eigenvalue combinations

3.2 The 8 commitment paths (candidate b.ii structural setup)

Within the fermion sublattice, the 8 commitment paths label by: - σ_BF eigenvalue: -1 (fixed at fermion sublattice; no choice) - $\hat{\gamma}^5$ eigenvalue: ±1 (L vs R Weyl spinor) - T_3^weak eigenvalue: ±1/2 (up vs down weak isospin) - Q̂ eigenvalue: ±1, 0 (electric charge — but discrete-eigenvalue set isn't binary)

So set (b.ii) has σ_BF fixed at -1 and gives only 2^2 = 4 paths from $\hat{\gamma}^5$ × T_3^weak. Not 8 paths.

Revised candidate (b.iii): {Q̂, T_3^weak, T_3^color} — three commuting Cartan generators with ±1-like eigenvalues - Q̂: ±1 (e or e^+ or quark charges in suitable units) - T_3^weak: ±1/2 (SU(2)_L) - T_3^color: r-related (substrate color triad)

But T_3^color has eigenvalues in {1/2, -1/2, 0} on color triplet → 3-state, not binary.

Refined candidate (b.iv): 3 binary CP-relevant generators selecting from {Q̂, P̂_op, T̂, Ĉ} (discrete symmetries) — but {Q̂, P̂_op} = 0 anti-commute (step 1).

The 8 commitment paths with one structurally forbidden via {Q̂, P̂_op} = 0 anti-commutation:

If commitment requires simultaneous specification of (Q̂-eigenvalue, P̂_op-eigenvalue, third-generator-eigenvalue), then {Q̂, P̂_op} = 0 anti-commutation FORBIDS simultaneous diagonalization. The path (Q = +1, P = +1, third = +1) is structurally inconsistent with (Q = -1, P = +1, third = +1) at the same time — anti-commuting operators can't have both eigenvalue specifications simultaneously.

Of the 8 paths labeled by (Q-sign, P-sign, χ-sign): some pairs are inconsistent under anti-commutation. The "forbidden" structure isn't 1 path out of 8 — it's a 2-state subset that's anti-commutator-forbidden.

v0.1 honest finding: candidate (b) needs sharpening. The simple "1 path out of 8 forbidden" reading does NOT cleanly map to {Q̂, P̂_op} = 0 anti-commutation. Multi-week v0.2+ work to identify the actual substrate-mechanism.

4. Alternative mechanism candidates

Per v0.4 Section 6.3, multiple candidate mechanisms:

4.1 Candidate (a) — Pin(2) Z_2 chirality × N_c × {±1}

3 binary choices: $\hat{\gamma}^5$ × color-triad-choice × sign - 2^3 = 8 paths - Forbidden path: chirality-color-sign combination violating color-singlet requirement - Substrate-mechanism: SU(3) color singlet requires antisymmetric combination; one of 8 combinations might be the symmetric (forbidden) state

Concrete: for 3-quark proton with N_c = 3 colors and 3 quarks, the color-singlet combination is $1/\sqrt{6}$ · (RGB - RBG + …). Of 3! = 6 color permutations, ONE specific antisymmetric combination is the singlet. NOT 1 out of 8.

Doesn't fit cleanly either.

4.2 Candidate (c) — 3 spatial commitment directions × {forward, backward}

3D space commitment × {±} = 2^3 = 8 paths. Forbidden path: spatial parity-violating commitment? But parity is conserved in EM + strong.

Doesn't have substrate-mechanism for the 1/8 redistribution.

4.3 Candidate (d) — Tsirelson^2 - S^2_BST factor decomposition

This is potentially the cleanest reading:

Tsirelson^2 = $2^{g/2}$(rank^2) = 128/16 normalized = 8 S^2_BST = (C_2 · N_c · g)/2^(rank^2) = 126/16 normalized = 7.875 Deviation = (2^g - C_2 · N_c · g) / 2^(rank^2) = 2/16 = 1/8

The "2" in the numerator IS the structural deviation: - 2 = N_max - 1 - N_c · (g + rank) - … NO, let me compute. 2^g - C_2·N_c·g = 128 - 126 = 2. - 2 = 2 (trivial) - Or: 2 = N_c - 1 = 3 - 1 = 2 (with N_c = 3) — this is the substrate's "off-by-N_c-minus-1" gap

Substantive structural reading: substrate Bell-CHSH max is HUA-LOOK-NORMALIZED to be 2 units below the qubit-Tsirelson max. The deviation is in the substrate-natural rank-1 projector restriction (only V_(0,0) accessible to B̂), which can't achieve the full 2-dim qubit Hilbert subspace required for Tsirelson saturation.

Mechanism: substrate has rank-1 access to Bell-CHSH eigenspace, NOT rank-2 (qubit) access. The 1-dim restriction reduces effective Bell value from Tsirelson to a substrate-natural maximum 2 units below (in normalized units 2^(rank^2) = 16).

This is cleaner than the 8-sided die metaphor: the substrate's 1/8 deviation is a 1-dim-vs-2-dim restriction, not a 1-out-of-8 forbidden path.

The 8 in "8-sided die" might be a coincidence: 8 = 2^N_c = 2^3 and 1/8 = 1/2^N_c both contain 2^N_c factor, but they arise from different structural sources (paths vs deviation normalization).

5. v0.1 honest conclusion

The structural identity Tsirelson^2 - S^2_BST = 1/2^N_c = 1/8 is RIGOROUSLY CLOSED (T2399 STRUCTURALLY VERIFIED EXACT), but the substrate-dynamical mechanism producing it remains FRAMEWORK-PLUS (Cal #132 step 9 tier).

Two readings of the mechanism:

(I) 8-sided die candidate (b): 2^N_c = 8 commitment paths with one structurally forbidden via {Q̂, P̂_op} = 0 anti-commutation. Cal #29 audit risk-flag: doesn't cleanly map to 1-path-out-of-8 structure.

(II) Rank-1 vs rank-2 substrate restriction (candidate d-revised): substrate Bell-CHSH operator is rank-1 projector onto V_(0,0) (T2399); CAN'T achieve qubit-Tsirelson which requires rank-2 access. The 2-unit normalized deviation (in 1/16 units) reflects substrate's rank-1 restriction. Cleaner structural mechanism.

v0.1 disposition: reading (II) is more substrate-natural and avoids the back-fit risk Cal #29 flagged for reading (I). v0.2+ multi-week formal derivation should pursue reading (II) and check whether the 8-path metaphor (reading I) corresponds to a structurally-distinct mechanism or is a coincidence.

6. Empirical discriminator (entropy vs constraint)

Per Casey 2026-05-26 AM directive (Track DC sub-question):

- ENTROPY: deviation environment-dependent; statistical fluctuation
- CONSTRAINT: deviation invariant; structural forbidden state

SP-30-1 Vienna Bell test measures Tsirelson^2 - S^2_observed across multiple experimental configurations. Predictions:

- If reading (II) rank-1 substrate restriction holds: deviation = 1/8 EXACT and invariant across all configurations (substrate's Bell-CHSH operator is fixed; doesn't depend on apparatus details) — CONSTRAINT
- If reading (I) 8-sided die structurally forbidden state: deviation = 1/8 EXACT and invariant — CONSTRAINT (same prediction; experimentally indistinguishable from reading II without theoretical work)
- If thermal/statistical mechanism: deviation environment-dependent — ENTROPY

Both reading (I) and (II) predict CONSTRAINT (invariant 1/8 deviation). Distinguishing them requires explicit theoretical derivation, not just experiment.

7. Substantive multi-week derivation path (v0.2+)

7.1 Pursue reading (II) rank-1 substrate restriction

Step 1: derive substrate's rank-1 Bell-CHSH access from A_sub structure (V_(0,0) ground state projection per step 9 vacuum kicker).

Step 2: derive why substrate-rank-1 specifically produces 2-unit normalized deviation (in 1/16 units) below qubit-rank-2 Tsirelson.

Step 3: substrate-mechanism for why qubit-rank-2 Hilbert subspace is required for Tsirelson saturation, contrasted with substrate-rank-1 restriction.

Step 4: explicit derivation: |Tsirelson - S_BST| as quantitative consequence of rank restriction; compare 1/8 numerical value.

Step 5: rigorous derivation of EXACT identity $2^g - C_2 \cdot N_c \cdot g = 2$ from BST primary integer structure.

7.2 Verify reading (I) 8-sided-die has substrate-natural correspondence

Multi-week: enumerate all 8 paths in candidate (b.iv) $\{\hat{Q}, \hat{P}_{op}$, third-generator$\}$ basis; identify whether $\{\hat{Q}, \hat{P}_{op}\} = 0$ anti-commutation eliminates EXACTLY ONE path; check if remaining 7 paths give the substrate-natural 1/8.

If yes: reading (I) is substrate-natural mechanism. If no: reading (I) is metaphorical, not structural.

7.3 Cal Thread 4 cold-read

Cold-read of Sections 4 + 5 mechanism candidates + Section 6 empirical discriminator. Type-check: rank-1 restriction (Type B algebraic) vs 8-path structure (potentially Type C level-crossing).

8. Honest scope (Cal #27 STANDING + Cal #29 STANDING)

What's RIGOROUSLY CLOSED / SVC: - T2399 EXACT identity $\text{Tsirelson}^2 - S^2\_BST = 1/2^{N_c} = 1/8$ (STRUCTURALLY VERIFIED) - $\{\hat{Q}, \hat{P}_{op}\} = 0$ (Cal #132 SVC) - $[\hat{B}, \hat{T}_{tick}]$ rank-1 vacuum kicker (Cal #132 FRAMEWORK-PLUS, step 9)

What's FRAMEWORK-PLUS in v0.1: - Section 3 candidate (b) 8-sided die: structural setup but Cal #29 audit flag risk (doesn't cleanly map to $\{\hat{Q}, \hat{P}_{op}\} = 0$ anti-commutation) - Section 4.3 candidate (d-revised) rank-1 substrate restriction: cleaner reading, multi-week explicit derivation needed

What's INTERPRETIVE in v0.1: - Both readings (I) and (II) predict CONSTRAINT (invariant 1/8 deviation) — experimentally indistinguishable from each other at SP-30-1 Vienna level - Step 1-5 multi-week derivation plan (Section 7.1)

What's NOT in v0.1: - Explicit numerical derivation of 1/8 from substrate-mechanism (multi-week) - Resolution of reading (I) vs reading (II) substrate-mechanism (multi-week) - Explicit rank-1 restriction quantitative consequence (multi-week)

Cal #27 STANDING reflexive trigger count: 2 triggers (Section 4.3 reading (II) feels substrate-natural; Section 5 "rank-1 vs rank-2" structural reading). Both flagged INTERPRETIVE in honest scope.

Cal #29 STANDING risk-flag: candidate (b) 8-sided die mechanism has back-fit risk; v0.1 explicitly flagged. Reading (II) is offered as cleaner substrate-natural alternative without back-fit risk.

9. Coordination

Cal: Thread 4 cold-read on candidate (b) 8-sided die back-fit risk + reading (II) rank-1 restriction structural cleanness. Type-check on multi-week explicit derivation path.

Elie: Toy 3538+ candidate — Cal #29 question-shape pre-audit required. Explicit substrate-mechanism for 1/8 deviation could be tested via numerical computation of substrate Bell-CHSH matrix elements at higher cutoff (Phase B 45-66 nodes).

Grace: catalog cross-references for Bell-CHSH-related observables; SP-30-1 Vienna Bell test cross-link to Track DC mechanism candidates.

Keeper: integration; Vol 15 Ch 9 case study draft includes Track DC v0.1 framework status as substantive multi-week Lyra-lane work.

— Lyra, Track DC v0.1 Bell 1/8 explicit derivation framework filed Tuesday 2026-05-26 ~10:18 EDT per Casey all-tracks authorization + Keeper Option 1 sequencing. FRAMEWORK-PLUS (Cal #132 step 9 tier; multi-week explicit derivation pending). Substantive v0.1 finding: candidate (b) 8-sided die has Cal #29 STANDING audit back-fit risk; reading (II) rank-1 substrate restriction is cleaner substrate-natural mechanism. Both readings predict CONSTRAINT (invariant 1/8) per SP-30-1 Vienna empirical discriminator; theoretical distinction requires v0.2+ multi-week formal derivation. Cal #27 STANDING applied 2 reflexive triggers; Cal #29 STANDING risk-flag preserved through.